Assignment 1

Module 3 – 11.02.2026

19-205-0810:OPERATIONS RESEARCH

Question 1: Transportation Problem (TP)

Problem Statement:

A company has **three warehouses** W1, W2, W3 with supplies of 30, 40, and 20 units, respectively. They need to supply **three retail stores** S1, S2, S3 with demands of 20, 50, and 20 units.

The transportation costs (₹ per unit) are given below:

Warehouse \ Store	S1	S2	S3
W1	8	6	10
W2	9	12	13
W3	14	9	16

Formulate this as a **Transportation Problem**.

Find an **initial basic feasible solution** using:
a) **Northwest Corner Method**
b) **Least Cost Method**

Compute the **total transportation cost** for each method.

Question 2: Assignment Problem (AP)

Problem Statement:

Four technicians need to be assigned to four machines. The cost (₹ per day) of assigning each technician to each machine is:

Technician \ Machine	M1	M2	M3	M4
T1	9	2	7	8
T2	6	4	3	7
T3	5	8	1	8
T4	7	6	9	4

Formulate this as an **Assignment Problem**.

Solve it using the **Hungarian Method** to find the **minimum total cost assignment**.

Verify the solution by computing the **total assignment cost**.

Convert this to a **maximization problem** if the numbers represent **profit per day**, and solve.

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